

This is a classic "Hello World" project for AgTech. It's satisfying because you get to see a physical result—a pump springing to life—based on environmental data. Since you have an Arduino, a sensor, and a pump, let's get this system running.

Project: The Thirsty Plant Rescuer

This project uses an Arduino to monitor the moisture level of soil. When the soil becomes too dry, the Arduino triggers a relay to turn on a water pump until the soil is sufficiently hydrated.

1. Component List

Beyond the basics mentioned in your task, you'll need a few small connectors to make it work safely:

- **Microcontroller:** Arduino Uno (or similar).
- **Sensor:** Capacitive or Resistive Soil Moisture Sensor.
- **Actuator:** 5V/12V DC Water Pump.
- **Power Switch:** 5V Relay Module (Crucial: Arduino pins can't power a pump directly without frying the board).
- **Power Source:** 9V battery or DC adapter for the pump.
- **Misc:** Jumper wires, breadboard, and a small water tube.

2. Wiring Schematic

Component	Arduino Pin	Note
Moisture Sensor VCC	5V	Power for the sensor
Moisture Sensor GND	GND	Ground
Moisture Sensor AO	A0	Analog signal for moisture level
Relay Module VCC	5V	Power for the relay
Relay Module GND	GND	Ground
Relay Module IN	Pin 8	Signal to trigger the pump

Warning: The pump should be wired through the Relay's **"NO" (Normally Open)** and **"COM" (Common)** terminals. The Arduino acts as the "brain" that flips the relay switch.

3. The Logic (Code)

Copy and paste this into your Arduino IDE. I've added comments to help you understand the automation logic.

```
// Pins
const int sensorPin = A0;
const int relayPin = 8;

// Calibration (Adjust these based on your specific soil)
const int dryThreshold = 800; // Higher value = drier soil
const int wetThreshold = 400; // Lower value = wetter soil

void setup() {
  pinMode(relayPin, OUTPUT);
```

```

    digitalWrite(relayPin, HIGH); // Keep pump OFF at start (Relays are
often active-low)
    Serial.begin(9600);
}

void loop() {
    int moistureValue = analogRead(sensorPin);
    Serial.print("Soil Moisture Level: ");
    Serial.println(moistureValue);

    if (moistureValue > dryThreshold) {
        // Soil is dry, turn pump ON
        Serial.println("Status: Soil is dry! Watering now...");
        digitalWrite(relayPin, LOW);
    }
    else if (moistureValue < wetThreshold) {
        // Soil is wet enough, turn pump OFF
        Serial.println("Status: Soil hydrated. Stopping pump.");
        digitalWrite(relayPin, HIGH);
    }

    delay(1000); // Wait 1 second before next reading
}

```

4. Step-by-Step Implementation

1. **Calibrate:** Put your sensor in a cup of dry soil and note the serial monitor value. Then, pour water in and note the new value. Update the dryThreshold in the code accordingly.
2. **The Plumbing:** Attach the tube to the pump and place the pump in a reservoir (water bottle/bowl). Ensure the tube is secured to the plant pot.
3. **The Test:** Insert the sensor into the dry soil. The relay should "click," and water should flow. Once the soil is damp, the pump should stop automatically.

Real-World Impact

- **Water Conservation:** Reduces waste by only watering when necessary.
- **Data Collection:** If you add an SD card module later, you can track how fast different plants "drink" water!

Do you have the relay module on hand, or are you planning to drive the pump using a transistor?